

Convert Adjacency matrix to Adjacency list

Solved

Difficulty: **Easy** Accuracy: **94.09%** Submissions: **1K+** Points: **2**

You are given an undirected graph represented by its adjacency matrix **mat**[[*i*][*j*], where **mat**[*i*][*j*] = **1** indicates that there is an edge between vertex *i* and vertex *j*, and **mat**[*i*][*j*] = **0** indicates no edge. Your task is to **convert** the given **adjacency matrix** into its corresponding **adjacency list** representation.

In the resulting adjacency list, each element **adj**[*i*] should contain all vertices that are directly connected to vertex *i*.

Examples :

Input: mat[[*i*][*j*] = [[0, 1, 0, 1],
[1, 0, 1, 0],
[0, 1, 0, 1],
[1, 0, 1, 0]]

Output: [[1, 3], [0, 2], [1, 3], [0, 2]]

Explanation:

The adjacency matrix represents the graph above and hence the adjacency list is [[1, 3], [0, 2], [1, 3], [0, 2]].

Input: mat[[*i*][*j*] = [[0, 1, 1, 0, 0],
[1, 0, 1, 0, 0],
[1, 1, 0, 1, 1],
[0, 0, 1, 0, 0],
[0, 0, 1, 0, 0]]

Output: [[1, 2], [0, 2], [0, 1, 3, 4], [2], [2]]

Explanation:

The adjacency matrix represents the graph above and hence the adjacency list is [[1, 2], [0, 2], [0, 1, 3, 4], [2], [2]].

Constraints:

$1 \leq V = \text{mat.size()} \leq 100$

$1 \leq \text{mat}[0].\text{size()} \leq 100$